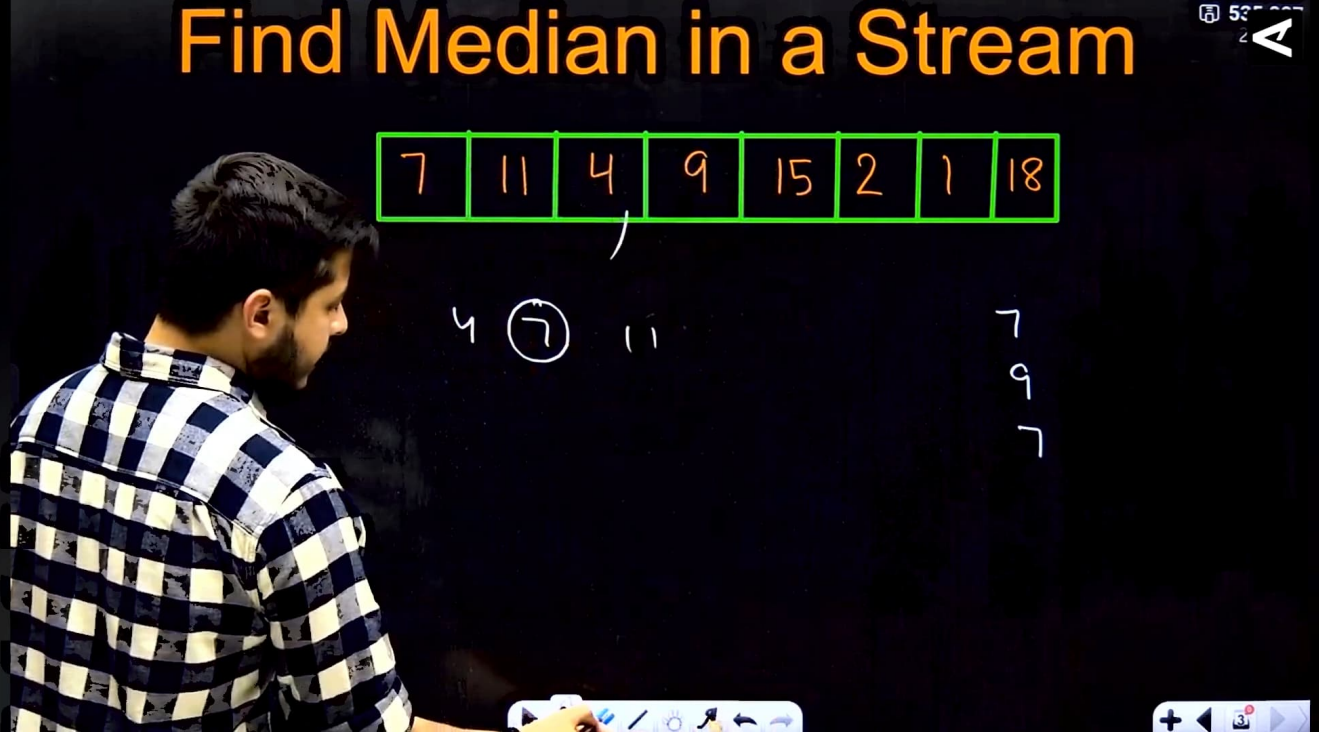
Find Median in a Stream



Find Median in a Stream

4 7 9 11 15

7
9
7
8
9

Find Median in a Stream

7	11	4	9	15	2	1	18

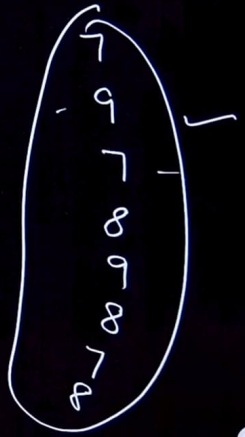
4 7 9 11 15 18

7 9 7 8 9 8 7 8

Find Median in a Stream

जी और उनको मैं उनकी सही पोजीशन पे लगा रहा हूं तो ऐसा

Find Median in a Stream



तो यहां पे क्या हो रहा है शिफ्टिंग हो रही है क्या हो रही

Find Median in a Stream

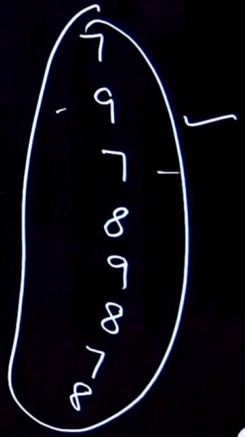
7	11	4	9	15	2	1	18

2	4	7	9	11	15

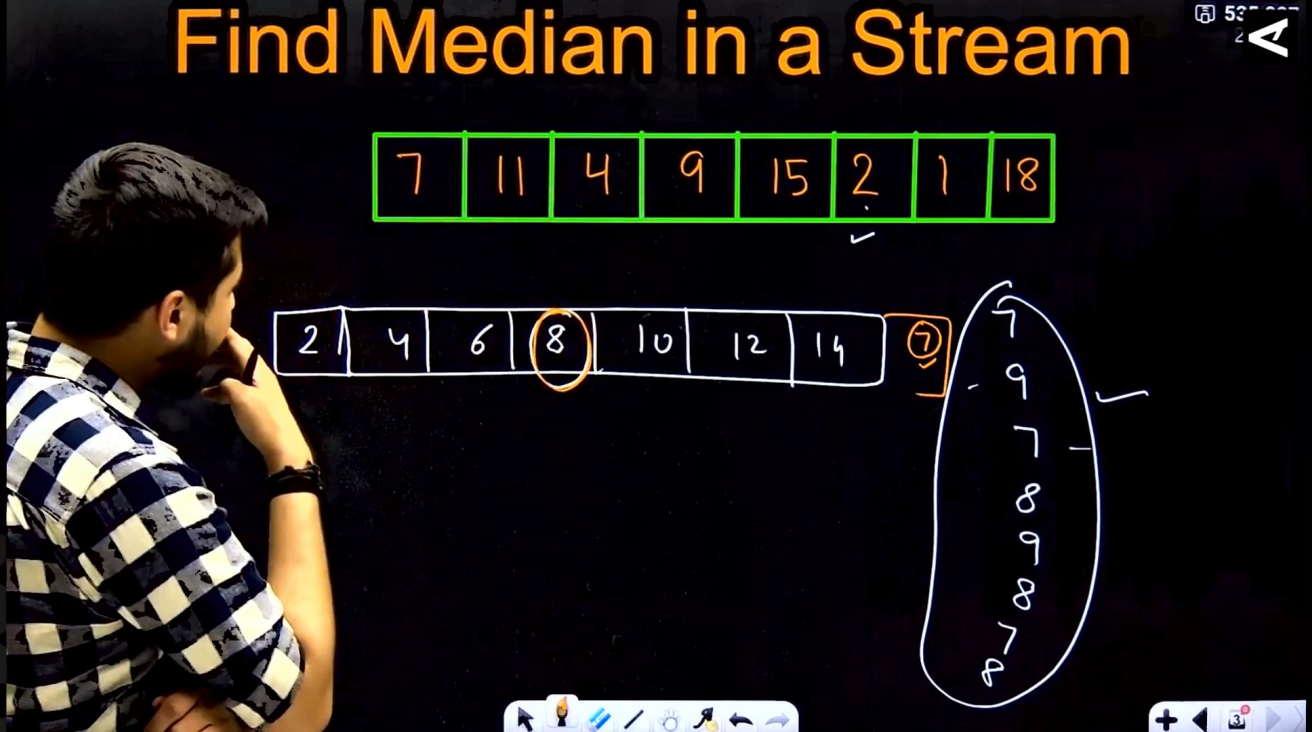
$0 + 1 + 2 + 3 \cdots n-1$

$\Downarrow$

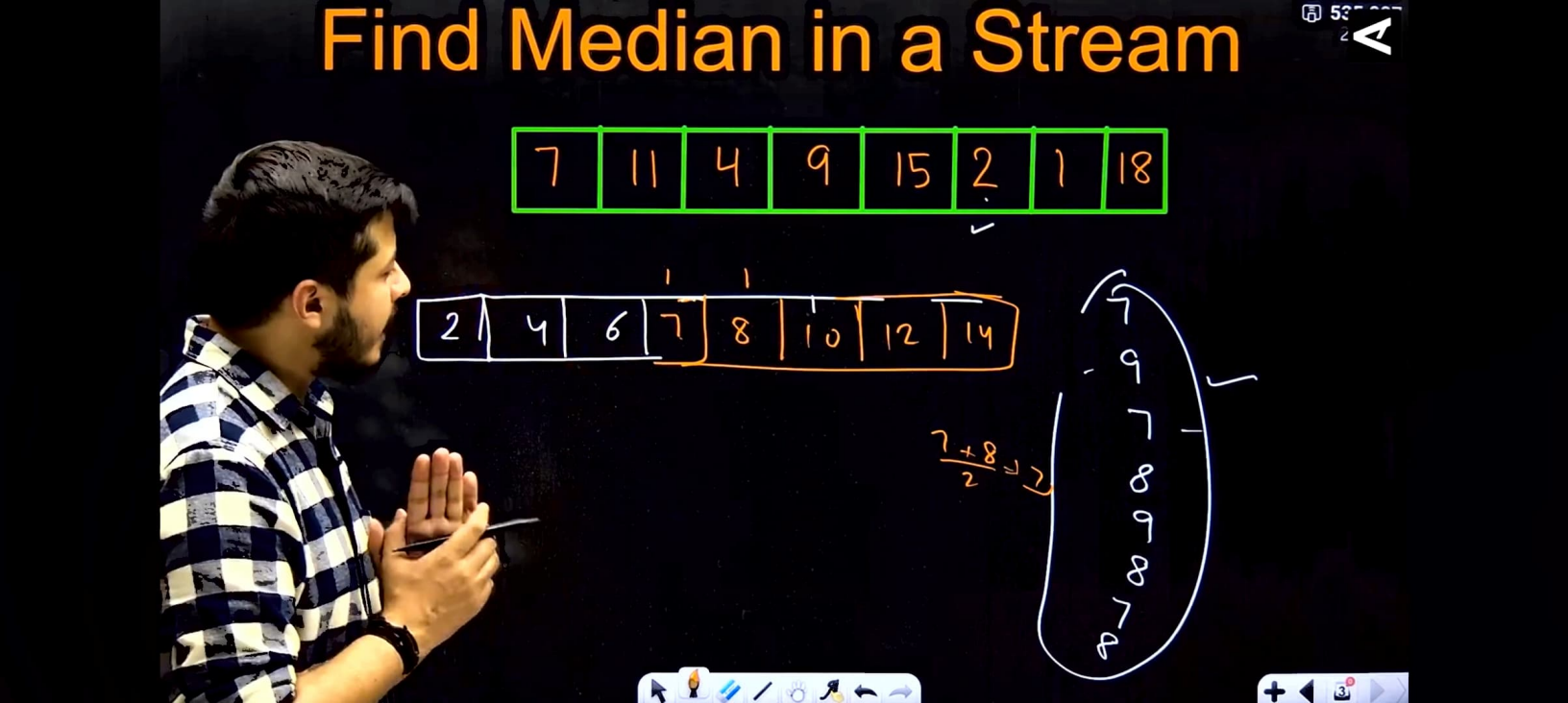
$O(N^2)$



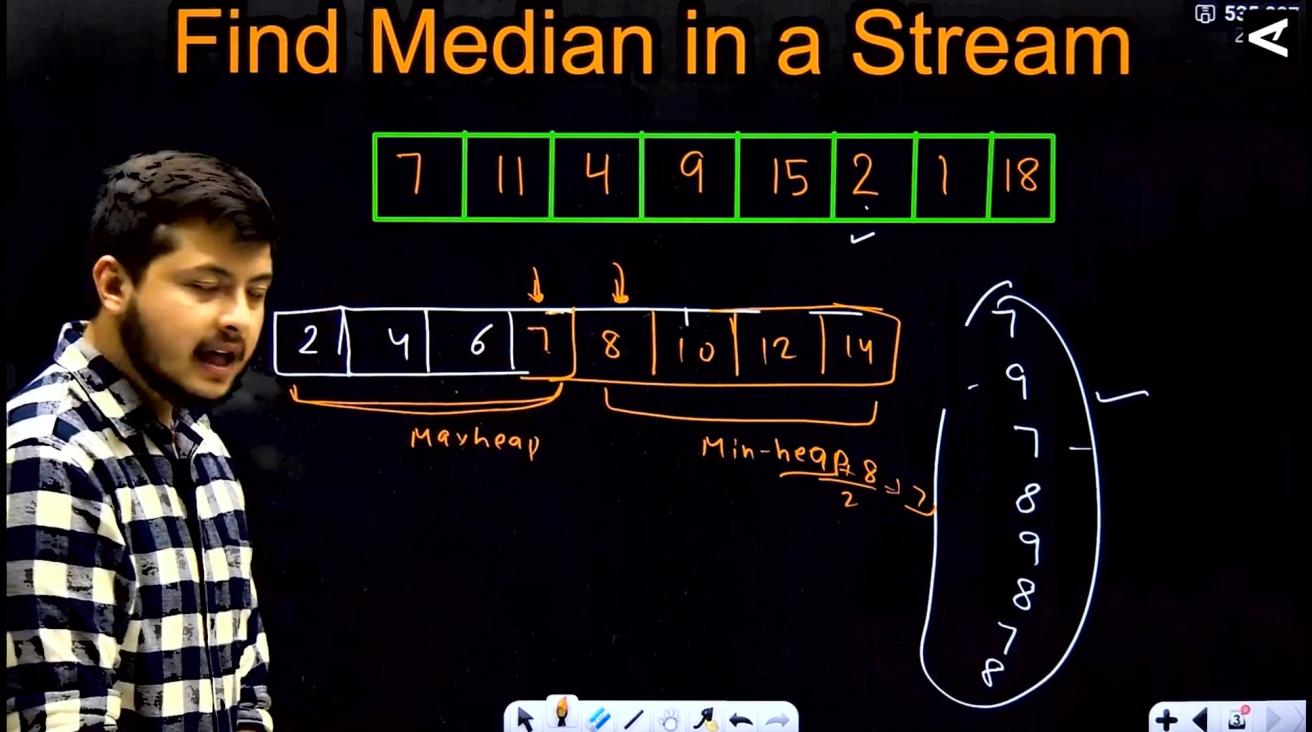
Find Median in a Stream



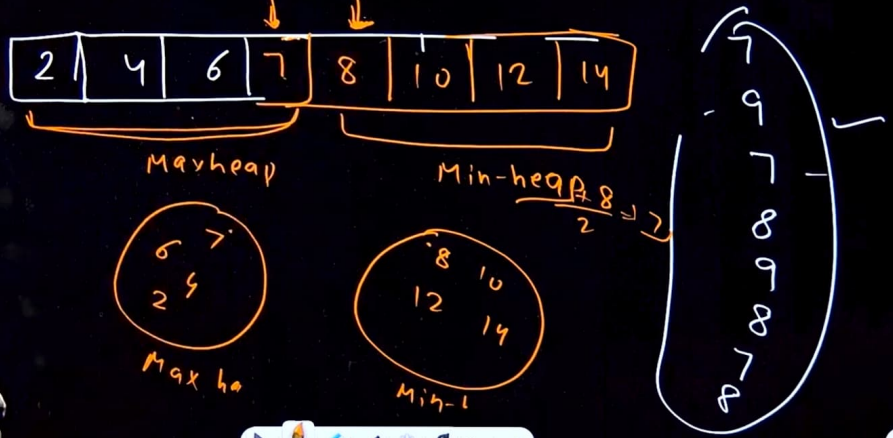
Find Median in a Stream



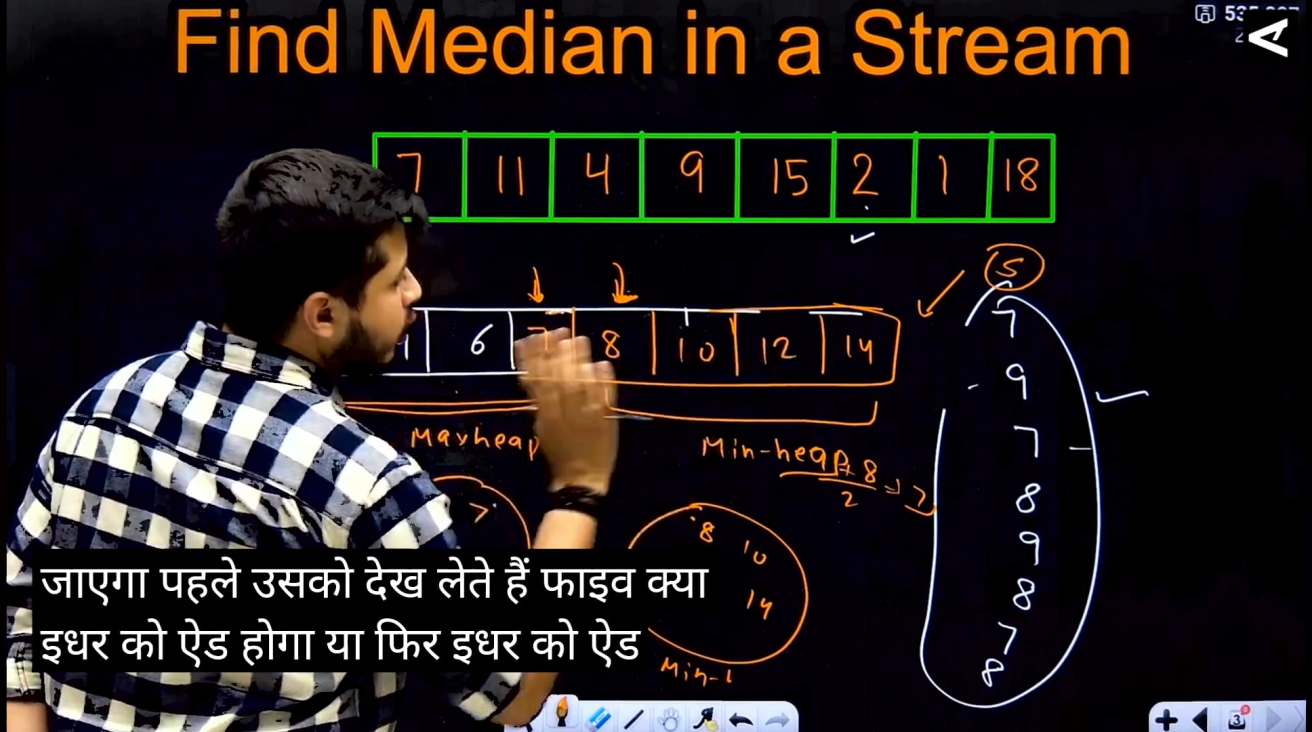
Find Median in a Stream



Find Median in a Stream

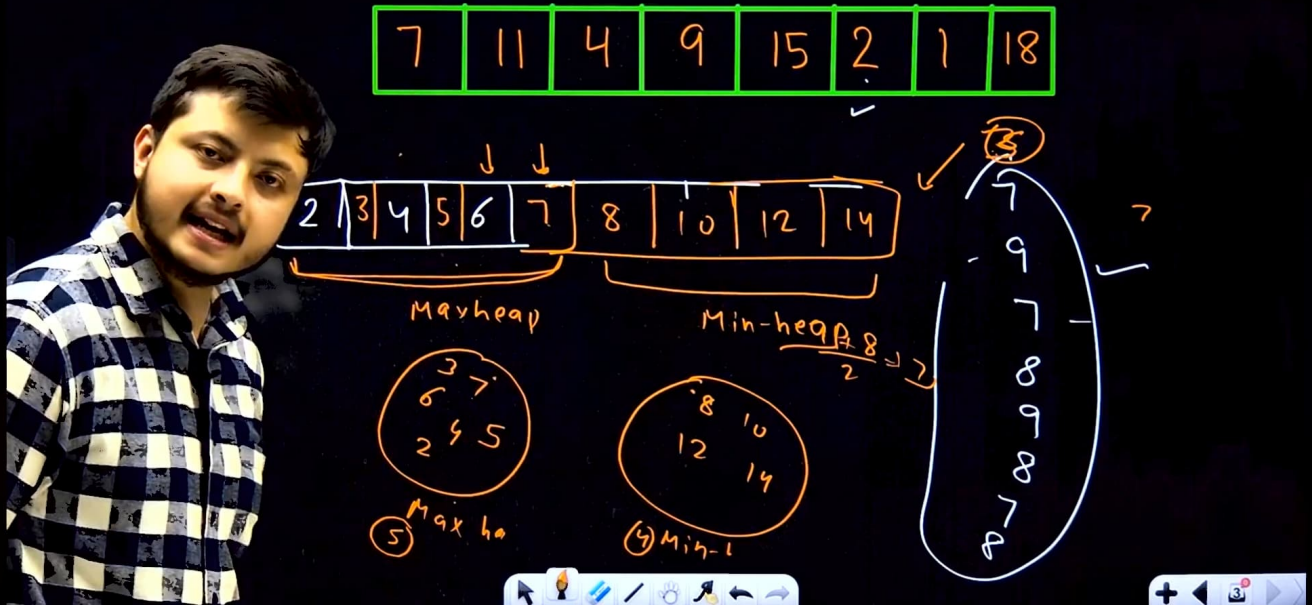


Find Median in a Stream

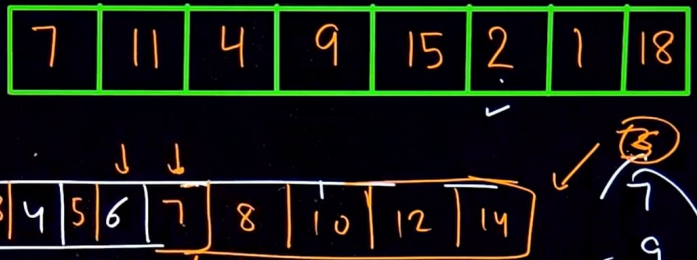


जाएगा पहले उसको देख लेते हैं फाइव क्या
इधर को ऐड होगा या फिर इधर को ऐड

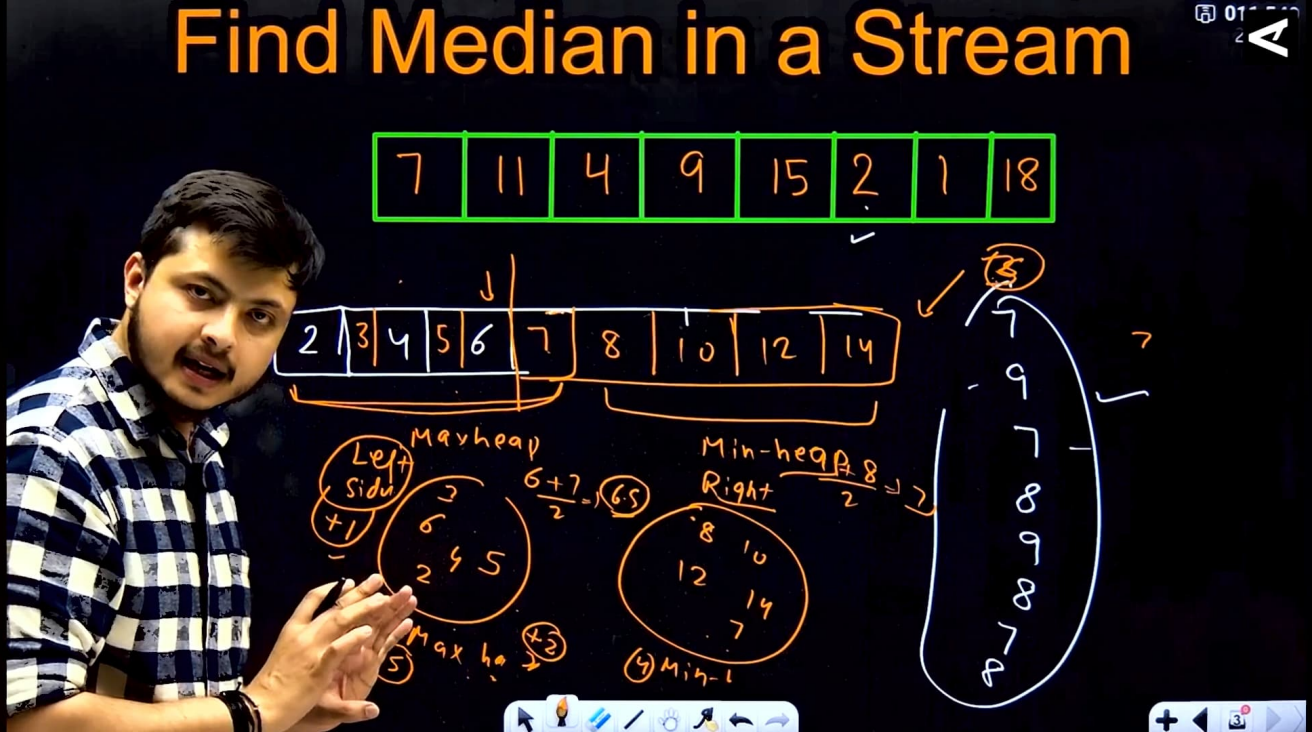
Find Median in a Stream

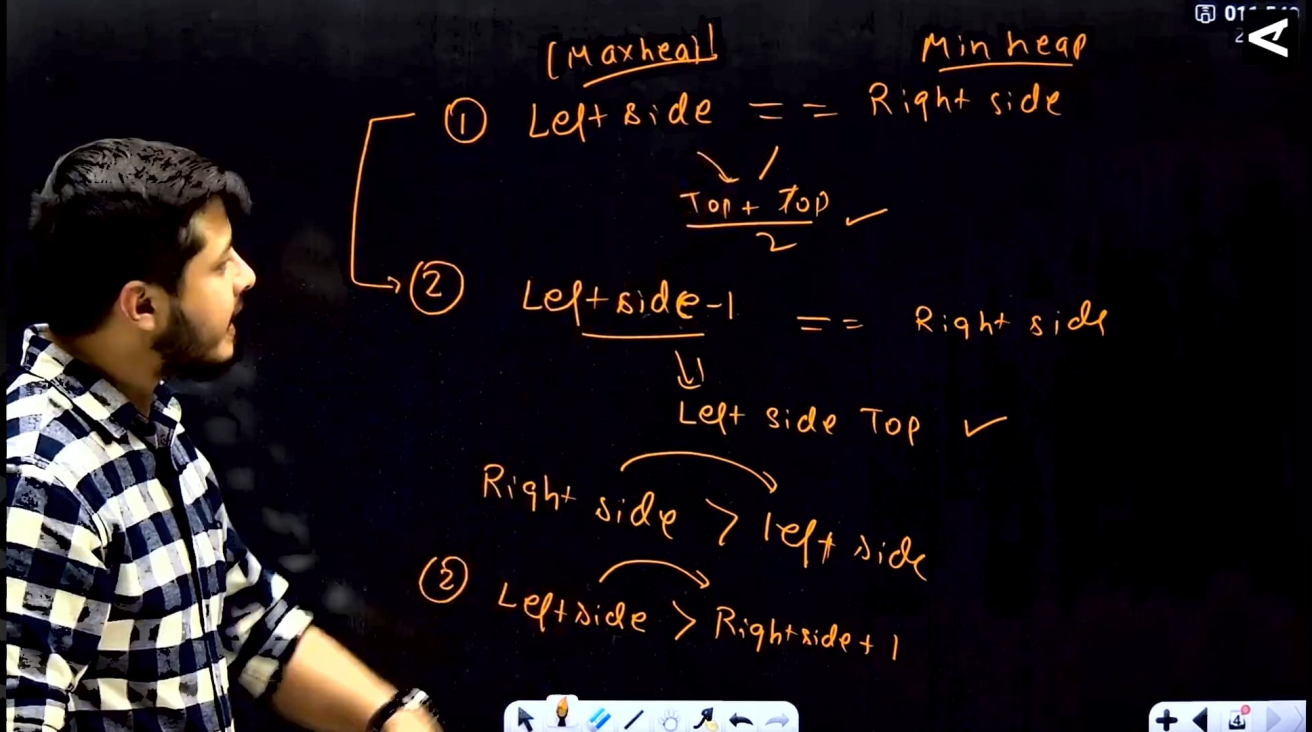


Find Median in a Stream



Find Median in a Stream




[Maxheap]
Min heap
① Left side == Right side
Top + Top
2
② Left side - 1 == Right side
Left side Top
Right side > left side
② Left side > Right side + 1

Problem | Editorial | Submissions | Comments

Find median in a stream

Hard Accuracy: 30.34% Submissions: 115K+ Points: 8

Given an input stream of **N** integers. The task is to insert these numbers into a new stream and find the median of the stream formed by each insertion of **X** to the new stream.

Example 1:

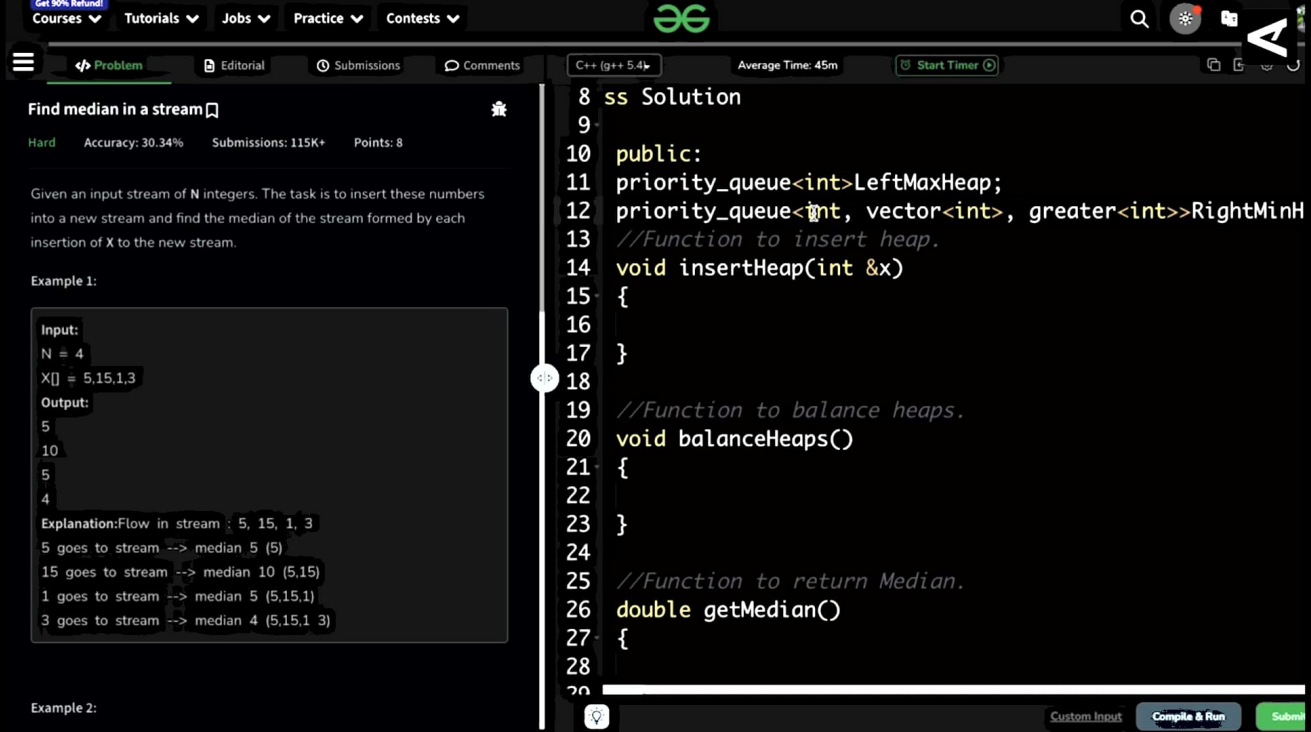
Input:
N = 4
X[] = 5,15,1,3
Output:
5
10
5
4
Explanation: Flow in stream : 5, 15, 1, 3
5 goes to stream --> median 5 (5)
15 goes to stream --> median 10 (5,15)
1 goes to stream --> median 5 (5,15,1)
3 goes to stream --> median 4 (5,15,1 3)

Example 2:

C++ (g++ 5.4) | Average Time: 45m | Start Timer

```
{
    public:

    //Function to insert heap.
    void insertHeap(int &x)
    {

    }

    //Function to balance heaps.
    void balanceHeaps()
    {

    }

    //Function to return Median.
    double getMedian()
    {

    }
};
```

Custom Input | Compile & Run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C++ (g++ 5.4)
Average Time: 45m
Start Timer
Find median in a stream
Hard
Accuracy: 30.34%
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Example 2:
ss Solution
public:
priority_queue<int>LeftMaxHeap;
priority_queue<int, vector<int>, greater<int>>RightMinH
//Function to insert heap.
void insertHeap(int &x)
{
}
//Function to balance heaps.
void balanceHeaps()
{
}
//Function to return Median.
double getMedian()
{
Custom Input
Compile & Run
Submi

10

Explanation: Flow in stream : 5, 10, 15
5 goes to stream --> median 5 (5)
10 goes to stream --> median 7.5 (5,10)
15 goes to stream --> median 10 (5,10,15)

Your Task:

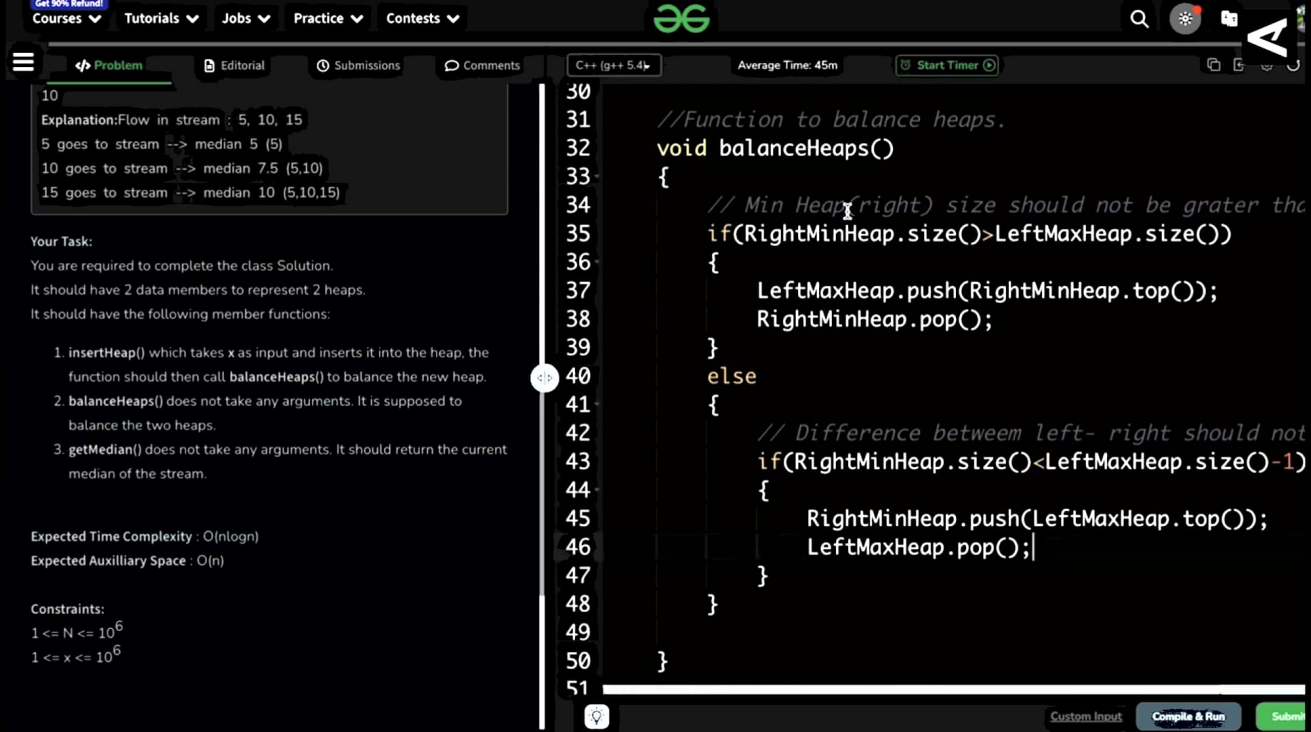
You are required to complete the class Solution.
It should have 2 data members to represent 2 heaps.
It should have the following member functions:

1. **insertHeap()** which takes **x** as input and inserts it into the heap, the function should then call **balanceHeaps()** to balance the new heap.
2. **balanceHeaps()** does not take any arguments. It is supposed to balance the two heaps.
3. **getMedian()** does not take any arguments. It should return the current median of the stream.

Expected Time Complexity : O(nlogn)
Expected Auxilliary Space : O(n)

Constraints:
$1 <= N <= 10^6$
$1 <= x <= 10^6$

```
17          if(LeftMaxHeap.empty())
18          {
19              LeftMaxHeap.push(x);
20              return;
21          }
22
23          if(x>LeftMaxHeap.top())
24          RightMinHeap.push(x);
25          else
26          LeftMaxHeap.push(x);
27
28          balanceHeaps;
29
30      }
31
32      //Function to balance heaps.
33      void balanceHeaps()
34      {
35
36      }
37
```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C++ (g++ 5.4)
Average Time: 45m
Start Timer
10
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Expected Time Complexity : O(nlogn)
Expected Auxilliary Space : O(n)
Constraints:
1 <= N <= 10^6
1 <= x <= 10^6
30
31 //Function to balance heaps.
32 void balanceHeaps()
33 {
34 // Min Heap(right) size should not be grater tha
35 if(RightMinHeap.size()>LeftMaxHeap.size())
36 {
37 LeftMaxHeap.push(RightMinHeap.top());
38 RightMinHeap.pop();
39 }
40 else
41 {
42 // Difference betweem left- right should not
43 if(RightMinHeap.size()<LeftMaxHeap.size()-1)
44 {
45 RightMinHeap.push(LeftMaxHeap.top());
46 LeftMaxHeap.pop();
47 }
48 }
49
50 }
51
Custom Input
Compile & Run
Submi

Problem Editorial Submissions Comments

10

Explanation: Flow in stream : 5, 10, 15
5 goes to stream --> median 5 (5)
10 goes to stream --> median 7.5 (5,10)
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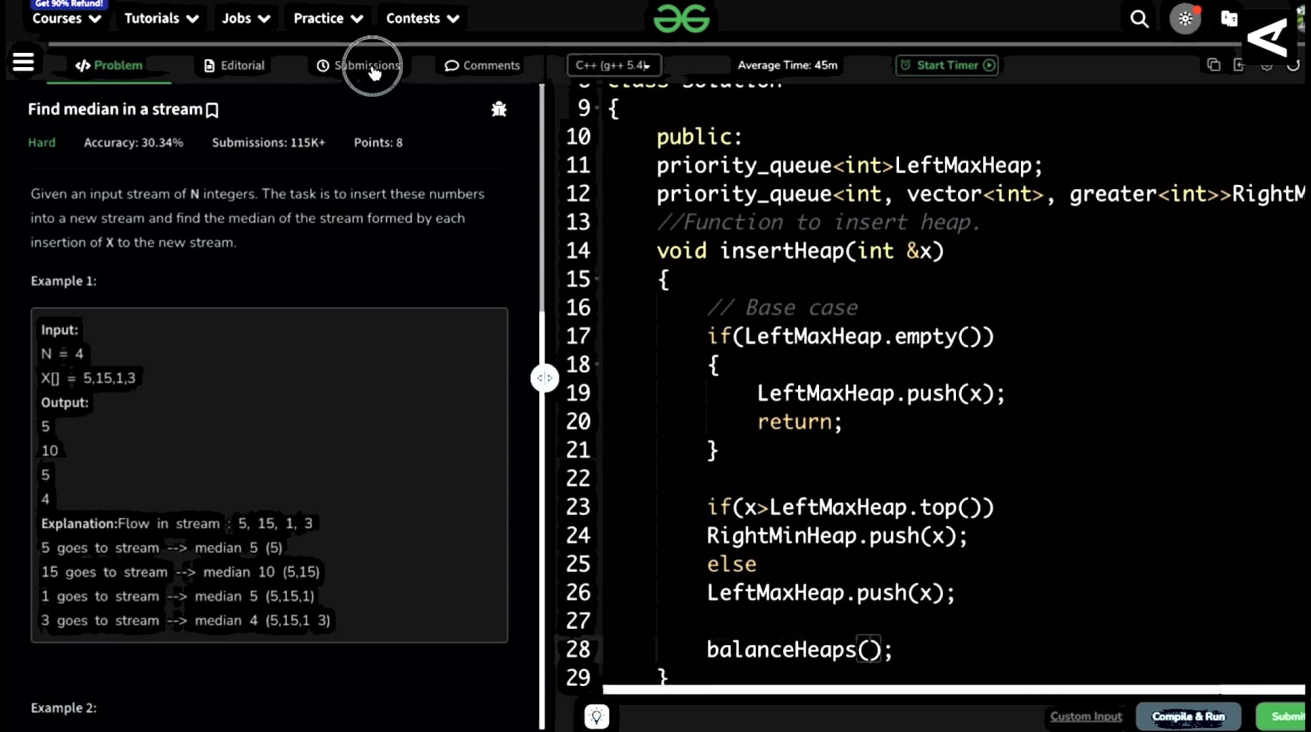
Expected Auxilliary Space : O(n)

Constraints:
$1 <= N <= 10^6$
$1 <= x <= 10^6$

C++ (g++ 5.4) Average Time: 45m Start Timer

```
46                  LeftMaxHeap.pop();
47              }
48          }
49
50      }
51
52      //Function to return Median.
53      double getMedian()
54      {
55          // Left > right
56          if(LeftMaxHeap.size()>RightMinHeap.size())
57          return LeftMaxHeap.top();
58          else
59          {
60              double ans = LeftMaxHeap.top()+ RightMinHeap
61              ans/=2;
62              return ans;
63          }
64      }
65  };
66
67
```

Custom Input Compile & Run Submi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5
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1 goes to stream --> median 5 (5,15,1)
3 goes to stream --> median 4 (5,15,1 3)
Example 2:
{
public:
priority_queue<int>LeftMaxHeap;
priority_queue<int, vector<int>, greater<int>>RightM
//Function to insert heap.
void insertHeap(int &x)
{
// Base case
if(LeftMaxHeap.empty())
{
LeftMaxHeap.push(x);
return;
}
if(x>LeftMaxHeap.top())
RightMinHeap.push(x);
else
LeftMaxHeap.push(x);
balanceHeaps();
}
Custom Input
Compile & Run
Submi

Find Median in a Stream.

Median – is the middle value in an ordered dataset.

for 7 11 4 => 4 [7] 11

& for 7 11 4 9

=> 4 [7 9] 11

$(7+9)/2 = 8$

7	11	4	9	15	2	1	18

Unordered	Ordered	Median
7	7	7
7 11	7 11	9
7 11 4	4 7 11	7
7 11 4 9	4 [7 9] 11	8
7 11 4 9 15	4 7 [9] 11 15	9

return this vector as ans.

1st: Brute-force Approach:

Either, with every coming element,

i) we sort the arr & find the median, T.C: $N(N \log N)$ ← for sorting

OR

ii) we place the new ele to it's correct position to maintain the sorted order.

$N * (N)$ ← for shifting

T.C: $O(N^2)$

FC $\simeq O(N^2 \log N)$

Approach 2:

We basically want our data to be in sorted form & when the new ele comes, we can simply insert into our DS without disturbing the sorted order.

2	4	6	8	10	12	14

Now if 7 comes

2	4	6	7	8	10	12	14

maxheap | minheap

if we observe carefully,

Median = $\frac{(\text{top ele (minh)} + \text{top ele (maxh)})}{2}$

The idea is to divide the data into 2 equal halves within minh & maxh in a way, their top element will be the median.

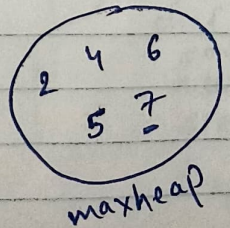
minheap.size() == maxheap().size() $7+8/2$

median = $\frac{\text{topEle(minh)} + \text{topEle(maxh)}}{2}$

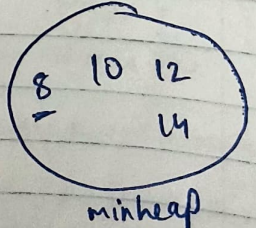
& if maxheap.size() > minheap.size or

maxheap.size() = minheap.size() + 1

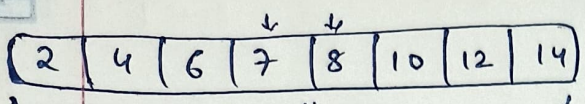
median = topEle(maxh). => 7



maxheap

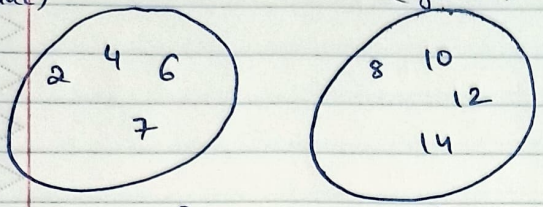


minheap



maxheap (leftside)

minheap (rightside)



maxheap minheap.

3 functions to complete.

```
//func to insert heap.
void insertHeap(int &x){
....
}
// function to balance heaps
void balanceHeaps(){
...
}
// function to return median
double getMedian(){
...
}
```

we have to call → this 2 will be called by compiler.

```
// Declare 2 heaps at the top as it will
// be used by all 3 funcs.
public:
priority_queue<int> LeftMaxHeap;
priority_queue<int, vector<int>, greater<int>> RightMinHeap;
```

Psuedocode:

① Leftside (Maxheap) == RightSide (Minheap)

median = (top + top) / 2

② Leftside - 1 == Rightside

median = Leftside (top)

③ Rightside > Leftside

pop from rightside (Minheap)

& push into leftside (Maxheap)

until

Leftside == rightside

or

Leftside - 1 == rightside

if (Leftside - 1 > Rightside)

pop from leftside

& push into rightside

```
// function to insert heap.
void insertHeap(int &x) {
    // Base case
    if (LeftMaxHeap.empty()){
        LeftMaxHeap.push(x);
        return;
    }
    if (x > LeftMaxHeap.top())
        RightMinHeap.push(x);
    else
        LeftMaxHeap.push(x);
    balanceHeaps();
}
```

```
//function to balance heaps.
void balanceHeaps(){
// MinHeap(Right) size should not be greater
// than MaxHeap
if (RightMinHeap.size() > LeftMaxHeap.size())
{ LeftMaxHeap.push(RightMinHeap.top());
  RightMinHeap.pop();
}
else{
// only if the difference b/w
// left-right > 1
if(RightMinHeap.size() < LeftMaxHeap.size()
                          -1)
{ RightMinHeap.push(LeftMaxHeap.top());
  LeftMaxHeap.pop();
}
}
```

left − right > 1	left	right
	5	4
	6	4
		X

```
//func to return Median
double getMedian(){
// Left > right
if (LeftMaxHeap.size() > RightMinHeap.size())
    return LeftMaxHeap.top();
else{
    double ans = top(MinH) + top(MaxH);
    ans /= 2;
    return ans;
}
}
```